A discriminant obstruction for fixed-point-free automorphisms of prime order of strongly regular graphs

Working note (draft, not reviewed)

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Status. This note generalizes the fixed-point-free part of D. Djematene's argument for $\mathrm{srg}(99,14,1,2)$ and order 3 (Zenodo, doi:10.5281/zenodo.23305123, unreviewed preprint). The proof below has been checked by its author and by one internal referee with an independent program, not by an external expert; the statement has been tested numerically on known graphs (Section 3). Nothing here is formalized. Section 4 lists consequences for open parameter sets; their novelty has not been fully established (Section 7).

1 Setting and statement

Let Γ be a connected $\mathrm{srg}(v,k,\lambda,\mu)$, not complete, with integral eigenvalues $k>r>s$ of multiplicities $1, f_r, f_s$. Its adjacency matrix satisfies $(A-rI)(A-sI)=\mu J$, and $(k-r)(k-s)=\mu v$. Let σ be an automorphism of prime order p without fixed points, so $v=pm$. Choose a representative x_U in each orbit ($U=1,\dots,m$) and put $A_b[U,V]=A(x_U,\sigma^b x_V)$ for $b\in\mathbb{Z}/p$. Then $A(\sigma^c x_U,\sigma^{c+b}x_V)=A_b[U,V]$, $A_b^{\mathsf T}=A_{-b}$, and A_0 has zero diagonal. Let ζ be a primitive p-th root of unity, $K=\mathbb{Q}(\zeta)$, $\mathcal{O}_K=\mathbb{Z}[\zeta]$, $K^+=\mathbb{Q}(\zeta+\zeta^{-1})$, $\pi=1-\zeta$, and

$$Q=\sum_b A_b\in M_m(\mathbb{Z}),\qquad M=\sum_b \zeta^b A_b\in M_m(\mathcal{O}_K).$$

Let $\rho:\mathcal{O}_K\to\mathbb{F}_p$ be reduction modulo π ($\rho(\zeta)=1$); then $\rho(\bar z)=\rho(z)$ and $\rho\circ\tau=\rho$ for every $\tau\in\mathrm{Gal}(K/\mathbb{Q})$. Write χ for the Legendre symbol modulo p.

For $\theta\in\{r,s\}$ let θ' be the other eigenvalue, $L_\theta=\mathbb{Z}^m\cap\ker(Q-\theta I)$ of rank a, $d=\det(B^{\mathsf T}B)$ for a basis B of L_θ, and, when $\mathcal{O}_K$ is a principal ideal domain ($p\le 19$), $L'_\theta=\mathcal{O}_K^m\cap\ker(M-\theta I)$ of rank a' and $d'=\det(B'^{*}B')$ for an $\mathcal{O}_K$-basis B'.

Theorem 1. *Assume $p\nmid(k-\theta)(\theta-\theta')$ and $p\le 19$. Then:*

(a) $a=a'$; consequently $f_\theta=pa$, and $\sum_U A_b[U,U]=(k-\theta')/p$ for every $b\neq 0$.

(b) d divides $((k-\theta)(\theta-\theta'))^a$; d' is a totally positive element of $\mathcal{O}_{K^+}$ dividing $(\theta-\theta')^{a'}=(\theta-\theta')^a$ in $\mathcal{O}_K$, so $\mathrm{N}_{K^+/\mathbb{Q}}(d')$ is a positive divisor of $|\theta-\theta'|^{a(p-1)/2}$.

(c) $\chi(d)\equiv\mathrm{N}_{K^+/\mathbb{Q}}(d')\pmod p$; in particular, since $p\nmid d$ by (b), $\mathrm{N}_{K^+/\mathbb{Q}}(d')\equiv\pm1\pmod p$.

(d) Let $g=\gcd(m,k-\theta)$. For every prime $\ell\mid m$ with $\ell\nmid(m/g)(\theta-\theta')$: $v_\ell(d)\equiv v_\ell(m)\pmod 2$.

Part (a) without (b)–(d) is a rank equality of the kind proved by Ishida (arXiv:2606.29183, Prop. 4.4) and holds for every prime p (the proof below works over the localizations $\mathbb{Z}_{(p)}$ and $\mathcal{O}_{K,(\pi)}$). Its numerical content is automatic from the parameters: $(k-\theta)(k-\theta')=\mu pm$ and $p\nmid k-\theta$ give $p\mid k-\theta'$. The new information is in (c) and (d).

2 Proof

Lemma 2. *Q is symmetric, $Q\mathbf{1} = k\mathbf{1}$ and $(Q - rI)(Q - sI) = p\mu J$. M is Hermitian, $(M - rI)(M - sI) = 0$, and $Q \equiv M \pmod{\pi}$ entrywise.*

Proof. For a matrix X on the vertices with $X(\sigma u, \sigma w) = X(u, w)$ put $X_b[U, V] = X(x_U, \sigma^b x_V)$ and $\Phi(X) = \sum_b X_b\, t^b \in M_m(\mathbb{Z}[t]/(t^p - 1))$. Invariance gives $(XY)_b = \sum_c X_c Y_{b-c}$, so Φ is a ring homomorphism with $\Phi(I) = I$ and $\Phi(J) = (1+t+\cdots+t^{p-1})J$. Apply Φ to $(A-rI)(A-sI) = \mu J$ and evaluate at $t = 1$ and $t = \zeta$. $Q^{\mathsf{T}} = Q$ and $M^* = M$ follow from $A_b^{\mathsf{T}} = A_{-b}$; $Q\mathbf{1} = k\mathbf{1}$ because row sums of Q are degrees; $Q - M = \sum_b (1 - \zeta^b)A_b$ and $\pi \mid 1 - \zeta^b$. □

Lemma 3. *Let R be a principal ideal domain and $X \in M_n(R)$. Then $L = \ker_R X$ is a direct summand of R^n: there are $B \in R^{n\times a}$ whose columns form a basis of L and $C \in R^{a\times n}$ with $CB = I_a$; and $BCY = Y$ for every matrix Y whose columns lie in L.*

Proof. $R^n/L \cong X(R^n)$ is finitely generated and torsion-free, hence free, so the sequence splits. If the columns of Y are Bc_j then $BCBc_j = Bc_j$. □

Step 1: two annihilating matrices. Let $N = (k - \theta)(Q - \theta' I) - p\mu J$ (integral, symmetric). Since $QJ = kJ$, $(Q-\theta I)N = (k-\theta)p\mu J - p\mu(k-\theta)J = 0$. If $Qx = \theta x$ then $k\,\mathbf{1}^{\mathsf{T}}x = \mathbf{1}^{\mathsf{T}}Qx = \theta\,\mathbf{1}^{\mathsf{T}}x$, so $Jx = 0$ and $Nx = (k - \theta)(\theta - \theta')x$. Let $N' = M - \theta' I$ (Hermitian); $(M - \theta I)N' = 0$ and $N'y = (\theta - \theta')y$ on $\ker(M - \theta I)$.

Step 2: Gram identities. With B, C from Lemma 3 for L_θ, put $G = B^{\mathsf{T}}B$ and $K = CNC^{\mathsf{T}}$. The columns of N lie in L_θ, so $BCN = N$; by symmetry $N = NC^{\mathsf{T}}B^{\mathsf{T}}$, hence $N = BKB^{\mathsf{T}}$. As $NB = (k - \theta)(\theta - \theta')B$,

$$GK = B^{\mathsf{T}}(BCN)C^{\mathsf{T}} = B^{\mathsf{T}}NC^{\mathsf{T}} = (k - \theta)(\theta - \theta')(CB)^{\mathsf{T}} = (k - \theta)(\theta - \theta')I_a.$$

So $d \det K = ((k - \theta)(\theta - \theta'))^a$. Likewise with B', C' for L'_θ, $G' = B'^*B'$, $K' = C'N'C'^*$: $N' = B'K'B'^*$ and $G'K' = (\theta - \theta')I_{a'}$, so $d' \det K' = (\theta - \theta')^{a'}$. G' and K' are Hermitian, so $d', \det K' \in \mathcal{O}_K \cap \mathbb{R} = \mathcal{O}_{K^+}$. For $\tau \in \mathrm{Gal}(K/\mathbb{Q})$, $\tau(G') = \tau(B')^*\tau(B')$ is the Gram matrix of a' linearly independent vectors of $\mathbb{C}^m$, hence positive definite: d' is totally positive. Taking norms gives (b).

Step 3: reduction modulo π. By Lemma 2, $\rho(M) = \rho(Q) =: \bar{Q}$, so $\bar{N} = \rho(k - \theta)(\bar{Q} - \theta' I) = \rho(k - \theta)\bar{N}'$. Hence, with $P = \bar{C}\bar{B}' \in \mathbb{F}_p^{a\times a'}$ and $P' = \bar{C}'\bar{B}$,

$$\bar{K} = \rho(k - \theta)\, P\bar{K}'P^{\mathsf{T}}, \qquad \bar{K}' = \rho(k - \theta)^{-1}\, P'\bar{K}P'^{\mathsf{T}}.$$

$\det K = ((k - \theta)(\theta - \theta'))^a/d$ is an integer whose prime factors divide $(k - \theta)(\theta - \theta')$, so $\bar{K}$ is invertible; $\rho(d')\rho(\det K') = \rho(\theta - \theta')^{a'} \neq 0$, so $\bar{K}'$ is invertible. The two factorizations give $a \le a'$ and $a' \le a$. Thus $a = a'$, P is invertible, and taking determinants, $\rho(k - \theta)^a\rho(\theta - \theta')^a\rho(d)^{-1} = \rho(k - \theta)^a \det(P)^2\rho(\theta - \theta')^a\rho(d')^{-1}$, i.e. $\rho(d') = \det(P)^2\rho(d)$. Hence $\chi(\rho(d')) = \chi(d)$. By Euler's criterion and $\rho \circ \tau = \rho$, $\chi(\rho(d')) = \rho(d')^{(p-1)/2} = \prod_{\tau\in\mathrm{Gal}(K^+/\mathbb{Q})} \rho(\tau d') = \rho(\mathrm{N}_{K^+/\mathbb{Q}}(d'))$. This is (c).

For (a): the isotypic decomposition of the θ-eigenspace of A under $\langle\sigma\rangle$ has a trivial part of dimension a and, for each of the $p - 1$ nontrivial characters, a part of dimension equal to the multiplicity of θ in the corresponding Galois conjugate of M, i.e. a'; so $f_\theta = a + (p - 1)a' = pa$. With $c_b = \sum_U A_b[U, U]$ ($c_0 = 0$, $c_b = c_{-b}$), $\operatorname{tr} N = a(k - \theta)(\theta - \theta')$ and $\operatorname{tr} N' = a(\theta - \theta')$ give $(k - \theta)(\operatorname{tr} Q - \operatorname{tr} M) = p\mu m$; applying the same to all conjugates $M_j = \sum_b \zeta^{jb}A_b$ shows that $\sum_b c_b\zeta^{jb}$ does not depend on $j \neq 0$, so $c_b = c$ for all $b \neq 0$, and $pc = p\mu m/(k - \theta) = k - \theta'$.

Step 4: gluing with 1. Q is symmetric and on $\mathbf{1}^\perp$ satisfies $(Q-rI)(Q-sI)=0$, so the k-eigenspace is $\mathbb{Q}\mathbf{1}$ and $W=\ker((Q-\theta I)(Q-kI))=\ker(Q-\theta I)\oplus\mathbb{Q}\mathbf{1}$. Let $L_W=\mathbb{Z}^m\cap W$ (rank $a+1$) with basis B_W, left inverse C_W and $G_W=B_W^{\mathsf{T}}B_W$. Put $\alpha=m/g$, $\beta=(k-\theta)/g$ and $N_W=\alpha(Q-\theta' I)-\beta J$. Then $(Q-\theta I)(Q-kI)N_W=0$ (because $(Q-\theta I)(Q-\theta' I)=p\mu J$ and $(Q-kI)J=0$), N_W is $\alpha(\theta-\theta')$ on $\ker(Q-\theta I)$, and $N_W\mathbf{1}=(\alpha(k-\theta')-\beta m)\mathbf{1}=\alpha(\theta-\theta')\mathbf{1}$ since $\beta m=\alpha(k-\theta)$. As in Step 2, $G_WK_W=\alpha(\theta-\theta')I_{a+1}$, so $\det G_W$ divides $(\alpha(\theta-\theta'))^{a+1}$. The columns of $[B\mid\mathbf{1}]$ lie in L_W, so $[B\mid\mathbf{1}]=B_WT$ with $T\in M_{a+1}(\mathbb{Z})$; since $B^{\mathsf{T}}\mathbf{1}=0$ (Step 1), comparing Gram matrices gives $\operatorname{diag}(G,m)=T^{\mathsf{T}}G_WT$, hence $m\,d=\det(T)^2\det G_W$. For $\ell\mid m$, $\ell\nmid\alpha(\theta-\theta')$ this gives (d). □

Corollary 4 (test used below)**.** *Under the hypotheses of Theorem 1, let S be the set of primes dividing $(k-\theta)(\theta-\theta')$. By (b) and (d), $\chi(d)=\prod_{\ell\in S}\chi(\ell)^{v_\ell(d)}$ where $v_\ell(d)\equiv v_\ell(m)$ for the primes of (d) and $v_\ell(d)$ is otherwise unconstrained (if $a=0$, $d=1$). By (b)–(c), $\chi(d)$ lies in the subgroup of $\mathbb{F}_p^\times$ generated by the primes dividing $\theta-\theta'$, intersected with $\{\pm1\}$. If the two sets of possible values are disjoint, or if some prime $\ell\mid m$ of (d) is not in S (so ℓ cannot divide d) while $v_\ell(m)$ is odd, then Γ has no fixed-point-free automorphism of order p.*

Remark 5. For $\mathrm{srg}(99,14,1,2)$, $p=3$, $\theta=3$: $S=\{7,11\}$, $m=33$, $g=11$, $\alpha(\theta-\theta')=21$, so $v_{11}(d)$ is odd and $\chi(d)=\chi(11)=-1$, while $\theta-\theta'=7\equiv1$ forces $\chi(d)=+1$. This is Djematene's contradiction.

3 Numerical validation

For triangular graphs $T(n)$ and their complements, with σ a product of p-cycles on $\{1,\dots,n\}$ fixing at most one point ($n\equiv0,1 \pmod p$; fixed-point-free on pairs), the script `obstruction.py` computes L_θ, L'_θ (through $\mathbb{Z}^{(p-1)m}$ and the trace form, whose determinant is $p^{(p-2)a'}\,\mathrm{N}_{K^+/\mathbb{Q}}(d')^2$), $a, a', d, \mathrm{N}(d')$, and checks (a)–(d). Cases with the hypothesis satisfied: $p=3$ ($n=7,9,10,15$), $p=5$ ($n=10,11,15$), $p=7$ ($n=14,15$), $p=11$ ($n=22,23$), $p=13$ ($n=13,14,27$), $p=17$ ($n=17,18$), $p=19$ ($n=19,20$), each with its complement: 36 applicable (Γ,p,θ) in all; every prediction holds, and Corollary 4 never excludes these graphs. When the hypothesis fails, $a\neq a'$ does occur. A second, independently written program (SymPy, Smith normal form) reproduces the numbers on several of these graphs and checks two further families: the Petersen graph with a 5-cycle ($a=a'=1$, $d=2$, $\mathrm{N}(d')=9$) and the collinearity graph of $GQ(3,3)$, $\mathrm{srg}(40,12,2,4)$, with a fixed-point-free automorphism of order 5 ($a=a'=3$, $d=288$, $\mathrm{N}(d')=5184$). With its own parser it also reproduces the list of Section 4 exactly.

Falsification test on existing parameter sets. Applied to the 1,076 parameter sets of the table for which a graph is known, Corollary 4 excludes a fixed-point-free automorphism of order p in a single case: $\mathrm{srg}(209,100,45,50)$ with $p=11$. The known graphs with these parameters are the descendants of the regular two-graph on 210 points containing the Sym(7) graph $\mathrm{srg}(210,99,48,45)$ of Klin et al. (as in the SageMath database). Rebuilt independently, all 210 descendants are isomorphic, and the descendant has an automorphism group of order 24 (computed with nauty), so it has no automorphism of order 11 at all: no conflict with the theorem. Other two-graphs on 210 points have not been examined.

4 Application to open parameter sets

Corollary 4 was applied to the 2,889 parameter sets marked open in Brouwer's table ($v\le1300$; snapshot of October 2026), 2,819 with integral eigenvalues, for every prime $p\mid v$. Part (a) excludes nothing. The test excludes a fixed-point-free automorphism of order p in 34 cases:

(v,k,λ,μ)	p	(v,k,λ,μ)	p
(88,27,6,9), (88,60,41,40)	11	(99,14,1,2)	3
(133,44,15,14)	7	(217,128,72,80)	7
(231,140,85,84)	11	(261,208,165,168)	3
(273,192,136,132)	7	(364,121,48,36)	7
(385,324,273,270)	11	(550,162,75,36), (550,486,429,432)	11
(561,60,9,6)	11	(616,240,89,96), (616,375,230,225)	11
(637,576,520,528)	7	(714,248,97,80)	3
(714,552,425,432)	7	(756,150,28,30), (756,605,484,484)	7
(780,410,229,200)	5	(826,132,26,20)	7
(924,142,20,22), (924,143,22,22)	7	(924,780,658,660), (924,781,660,660)	7
(945,64,8,4), (945,880,819,825)	7	(950,312,115,96)	19
(1045,324,123,90)	11	(1085,256,48,64)	7
(1148,666,379,396)	7	(1177,972,801,810)	11
(1248,1044,875,864)	13		

Corollary 6 (conditional on the literature check)**.** *An* $\mathrm{srg}(88,27,6,9)$ *has no automorphism of order* 11.

Proof. An automorphism of order 11 has a number of fixed points $f \equiv 88 \equiv 0 \pmod{11}$. Behbahani and Lam (thesis 2009, Table 16) list $f \in \{0,11,22\}$ as the candidates and find no orbit matrix for $f = 11$ and $f = 22$. So $f = 0$, which Corollary 4 excludes: for $\theta = 3$, $\theta' = -6$, $m = 8$, $k - \theta = 24$, $g = 8$, $\alpha(\theta - \theta') = 9$, so $v_2(d) \equiv v_2(8) = 3$ is odd; with $\chi(2) = -1$ and $\chi(3) = 1$ modulo 11 this forces $\chi(d) = -1$, while $\theta - \theta' = 9$ and 3 generates a subgroup of $\mathbb{F}_{11}^{\times}$ of order 5 force $\chi(d) = +1$. □

5 Automorphisms with fixed points

Let now σ have prime order p, fixed set F ($f = |F|$, induced adjacency matrix A_F) and m regular orbits, so $v = pm + f$. Put $a[U,t] = A(x_U, t)$ for $t \in F$. On σ-invariant vectors (one coordinate per orbit, one per fixed vertex) A acts by

$$\tilde{A} = \begin{pmatrix} Q & a \\ p\,a^{\mathsf{T}} & A_F \end{pmatrix}, \qquad \tilde{A}\mathbf{1} = k\mathbf{1}, \qquad (\tilde{A} - rI)(\tilde{A} - sI) = \mu\,\mathbf{1}w^{\mathsf{T}}, \quad w = (p,\dots,p,1,\dots,1),$$

and $\tilde{A}$ is self-adjoint for the form $D = \mathrm{diag}(pI_m, I_f)$ induced by $\mathbb{Z}^v$. On the other isotypic components the fixed vertices carry no weight (their coordinates are σ-invariant), the paths through F and the term in J are multiplied by $\sum_b \zeta^b = 0$, and $M = \sum_b \zeta^b A_b$ still satisfies $(M - rI)(M - sI) = 0$, $M^* = M$, $M \equiv Q \pmod{\pi}$.

Theorem 7. *Let* $p \le 19$, $\theta \in \{r,s\}$ *with* $p \nmid c := (k-\theta)(\theta - \theta')$, *and assume*

$$\text{(F)} \qquad \det\big((k-\theta)(\theta I_f - A_F) + \mu J_f\big) \not\equiv 0 \pmod{p}.$$

Let $L = \mathbb{Z}^{m+f} \cap \ker(\tilde{A} - \theta I)$ *(rank* a*) with basis* B, *and* L', a', d' *as before. Then every vector of* L *has its fixed coordinates divisible by* p, *so* $B^{\mathsf{T}}DB = p\tilde{G}$ *with* $\tilde{G}$ *integral; with* $\tilde{d} = \det\tilde{G}$*: (a)* $a = a'$, *hence* $p \mid f_\theta$*; (b)* $\tilde{d} \mid c^a$*; (c)* $\chi(\tilde{d}) \equiv \mathrm{N}_{K^+/\mathbb{Q}}(d') \pmod{p}$*; (d) for every prime* $\ell \neq p$ *with* $\ell \mid v$ *and* $\ell \nmid \alpha(\theta - \theta')$, $\alpha = v/\gcd(v, k-\theta)$*:* $v_\ell(\tilde{d}) \equiv v_\ell(v) \pmod 2$.

Proof. Let $N = (k-\theta)(\tilde{A} - \theta' I) - \mu\mathbf{1}w^{\mathsf{T}}$. As before $(\tilde{A} - \theta I)N = 0$ and $Nx = cx$ on L (for such x, $w^{\mathsf{T}}x = \langle \mathbf{1}, x\rangle_D = 0$). The rows of N indexed by $t \in F$ are $(k-\theta)p\,a[U,t] - \mu p \equiv 0$ on the orbit columns and $R = (k-\theta)(A_F - \theta' I) - \mu J_f$ on the fixed columns, so $c\,x_F \equiv R x_F \pmod{p}$, i.e. $\big((k-\theta)(\theta I - A_F) + \mu J_f\big)x_F \equiv 0$, and (F) gives $x_F \equiv 0$. Put $X = N\,\mathrm{diag}(I_m, pI_f) = pND^{-1}$: it is integral, symmetric (because DN is), with columns in L, so $X = BKB^{\mathsf{T}}$ with $K = CXC^{\mathsf{T}}$, and $B^{\mathsf{T}}DX = p\,c\,B^{\mathsf{T}}$ gives $\tilde{G}K = cI_a$. Modulo p, $\bar{B} = \binom{\bar{B}_1}{0}$, $\bar{C}_1\bar{B}_1 = I$ and

$\bar{X} = \operatorname{diag}\big(\rho(k-\theta)(\bar{Q}-\theta' I), 0\big)$, so $\bar{K} = \rho(k-\theta)\,\bar{C}_1(\bar{Q}-\theta' I)\bar{C}_1^{\mathsf{T}}$ and $\rho(k-\theta)(\bar{Q}-\theta' I) = \bar{B}_1\bar{K}\bar{B}_1^{\mathsf{T}}$; Steps 3 of Theorem 1 apply verbatim with $(B_1, C_1, \tilde{d})$ in place of (B, C, d). For (d) use $W = \ker((\tilde{A}-\theta I)(\tilde{A}-kI)) = \ker(\tilde{A}-\theta I) \oplus \mathbb{Q}\mathbf{1}$, $N_W = \alpha(\tilde{A}-\theta' I) - \beta\mathbf{1}w^{\mathsf{T}}$ with $\beta v = \alpha(k-\theta)$, which is $\alpha(\theta-\theta')$ on W; the same computation gives $G_W K_W = p\alpha(\theta-\theta')I$, and the D-Gram matrix of $[B \mid \mathbf{1}]$ is $\operatorname{diag}(p\tilde{G}, v)$, so $p^a \tilde{d}\, v = \det(T)^2 \det G_W$. □

For $f = 0$ condition (F) is empty and Theorem 7 is Theorem 1. Condition (F) is sufficient, not necessary, for the divisibility of the fixed coordinates (in $P(25)$ with $\theta = -3$, $\tilde{d}$ is integral although (F) fails). Since (a)–(d) do not involve f or A_F, the arithmetic test of Corollary 4 (with v in place of m in (d)) gives: *if the test fails for $(v, k, \lambda, \mu, p, \theta)$, every automorphism of order p has a fixed subgraph with* $\det((k-\theta)(\theta I - A_F) + \mu J_f) \equiv 0 \pmod{p}$; for $f = 1$ this reads $(k-\theta)\theta + \mu \equiv 0$. The same holds when $p \nmid f_\theta$ (by (a)); that consequence is of the rank-equality type and probably contained in Ishida's work.

Lemma 8. *Let $p > \max(\lambda, \mu)$ and $\sigma \neq 1$ of order p. Then two fixed vertices have only fixed common neighbours, so Γ_F has exactly λ common neighbours in F for adjacent and μ for non-adjacent pairs. If $\mu \geq 1$ and $f \geq 2$, Γ_F is connected, non-adjacent vertices of Γ_F have equal degrees, and Γ_F is the join of the regular graphs induced on the connected components of its complement. If $\mu \geq 2$, then $f \leq v/(p+1)$.*

Proof. A non-fixed common neighbour of two fixed vertices brings its whole orbit, $p > \max(\lambda, \mu)$ common neighbours. For non-adjacent $x, y \in F$, counting edges between $N_F(x)$ and $N_F(y)$ from both sides gives $\mu\lambda + (\deg x - \mu)\mu = \mu\lambda + (\deg y - \mu)\mu$. If $\mu \geq 2$ and a fixed vertex x had all its neighbours fixed, every vertex at distance 2 from x would have two fixed neighbours, hence be fixed, and $\sigma = 1$; so every fixed vertex has a non-fixed neighbour, hence $\geq p$ of them, while every non-fixed vertex has at most one fixed neighbour: $pf \leq v - f$. □

With Lemma 8, the admissible fixed subgraphs for $p > \max(\lambda, \mu)$, $\mu \geq 2$ are $K_{\lambda+2}$, primitive $\operatorname{srg}(f, k_F, \lambda, \mu)$, and joins of single vertices, edgeless graphs, unions of cliques and primitive strongly regular graphs with the induced parameters; for each, (F) is computed from the spectrum (`scan_fixed3.py`). Theorem 7 was checked on Paley graphs $P(25)$, $P(49)$ ($p = 3$) and $P(81)$ ($p = 5$) with multiplication by a square of order p ($f = 1$), and on $T(n)$ and complements with p-cycles fixing 2–6 points ($p = 5, 7, 17$): in all 16 cases where (F) holds, every prediction holds; in the 10 cases where the parameter test fails, (F) indeed fails.

Corollary 9 (candidates; see Section 7)**.** *Combining Theorem 7, Lemma 8 and the arithmetic test (no search over graphs; the list of admissible fixed subgraphs is a parameter computation done by `scan_fixed3.py` and should be checked by hand):*

- $\operatorname{srg}(88, 27, 6, 9)$ *has no automorphism of order* 11 *($f \leq 7$, so $f = 0$);*
- $\operatorname{srg}(99, 14, 1, 2)$ *has no automorphism of order* 3 *($\Gamma_F \in \{\emptyset, K_3, \operatorname{srg}(9, 4, 1, 2)\}$, all excluded with $\theta = 3$);*
- $\operatorname{srg}(342, 33, 4, 3)$ *has no automorphism of order* 7 *($\Gamma_F = K_6$, excluded with $\theta = -5$);*
- $\operatorname{srg}(364, 33, 2, 3)$ *has no automorphism of order* 5 *($\Gamma_F = K_4$, excluded with $\theta = 5$);*
- $\operatorname{srg}(561, 60, 9, 6)$ *has no automorphism of order* 11 *($\Gamma_F \in \{\emptyset, K_{11}\}$, excluded with $\theta = 9$);*
- $\operatorname{srg}(650, 55, 0, 5)$ *has no automorphism of order* 11 *($f = 1$, excluded with $\theta = 5$ and with $\theta = -10$);*

and the same for the complementary parameter sets, since $\operatorname{Aut}(\Gamma) = \operatorname{Aut}(\bar{\Gamma})$ *(the hypothesis $p > \max(\lambda, \mu)$ of Lemma 8 is used for the listed parameter set, not for its complement). The second item is Djematene's theorem (fixed-point-free and triangle cases). For several values of f the exclusion comes from the nonexistence of any admissible fixed subgraph rather than from (F).*

6 Regular graphs with more eigenvalues

Nothing above uses that Γ has only three eigenvalues except through the identity $(A - rI)(A - sI) = \mu J$. Let Γ be a connected k-regular graph with distinct eigenvalues $k = \theta_0 > \theta_1 > \cdots > \theta_d$ and Hoffman polynomial $H(x) = \prod_{i \geq 1}(x - \theta_i) \in \mathbb{Z}[x]$, so that $H(A) = cJ$ with $c = H(k)/v \in \mathbb{Z}$ (an entry of the integer matrix $H(A)$). For an integral eigenvalue $\theta = \theta_j$ put $P_\theta(x) = H(x)/(x - \theta) \in \mathbb{Z}[x]$ and $c_\theta = (k - \theta)P_\theta(\theta)$. Replace N by $(k - \theta)P_\theta(\tilde{A}) - c\,\mathbf{1}w^{\mathsf{T}}$, N' by $P_\theta(M)$ and N_W by $\alpha P_\theta(\tilde{A}) - \beta \mathbf{1}w^{\mathsf{T}}$ with $\beta v = \alpha(P_\theta(k) - P_\theta(\theta))$. The bottom-left block of every power of $\tilde{A}$ is divisible by p, so the rows of N at fixed vertices reduce to $(k-\theta)P_\theta(A_F) - cJ_f$ on the fixed columns. The proofs go through word for word and give:

Theorem 10. *Let $p \leq 19$, θ an integral eigenvalue $\neq k$ with $p \nmid c_\theta$, and assume* $\det\big((k - \theta)(P_\theta(\theta)I - P_\theta(A_F)) + cJ_f\big) \not\equiv 0 \pmod{p}$. *Then (a)–(d) of Theorem 7 hold with $c = c_\theta$, with $\theta - \theta'$ replaced by $P_\theta(\theta)$, and with $\alpha = v/\gcd(v, P_\theta(k) - P_\theta(\theta))$.*

For strongly regular graphs this is Theorem 7. `general.py` checks Theorem 10 on Johnson graphs $J(n, 3)$ and Kneser graphs $K(n, 3)$ (four distinct eigenvalues) with products of p-cycles, with and without fixed points; all predictions hold in the cases computed so far. Applied to the 1,514 feasible intersection arrays of the BCN tables (Brouwer's machine-readable files, which carry no existence status), the parameter test gives very little: one exclusion of a fixed-point-free automorphism, of order 7 for $\{65, 44, 11; 1, 4, 55\}$ ($v = 924$), and one exclusion of an automorphism of order 11 with a single fixed point for $\{11, 10, 4; 1, 1, 5\}$ ($v = 210$). Neither array appears among the known graphs of the diameter-3 index of DistanceRegular.org; their existence status has not been checked.

7 Literature status

Behbahani–Lam (2011) list $2, 3, 5, 11$ as the possible primes for $\mathrm{srg}(88, 27, 6, 9)$, with the fixed-point-free order-11 case unresolved (5 orbit matrices). Efimov–Makhnev, *On automorphisms of a strongly regular graph with parameters (88,27,6,9)*, Dokl. Math. 86 (2012) 488–491, has not been accessed. Kazarina–Makhnev, *On automorphisms of strongly regular graphs with $\lambda = 2$ and $\mu = 3$*, Izv. UGTU 17 (2005) 174–194, probably treats $\mathrm{srg}(364, 33, 2, 3)$ and has not been accessed. No work on automorphisms of $\mathrm{srg}(342, 33, 4, 3)$, $\mathrm{srg}(561, 60, 9, 6)$ or $\mathrm{srg}(650, 55, 0, 5)$ was found in a first search. Ishida (arXiv:2606.29183, §8) treats other parameter sets. Until these sources are read, the items of Corollary 9 other than the second are candidates, not established new results.